

LSGI3262 Laser Scanning Lab 3

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Task 1 - Compare different settings and methods of denoising

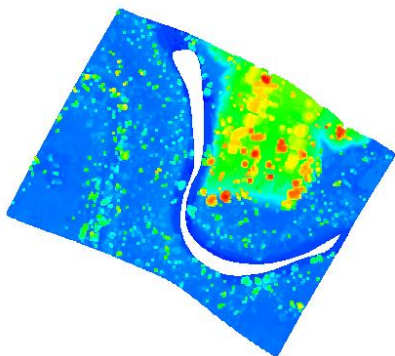
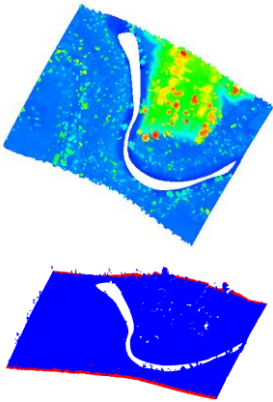
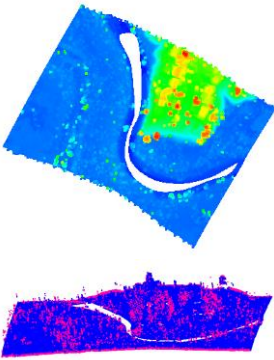
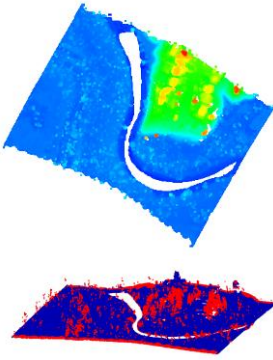
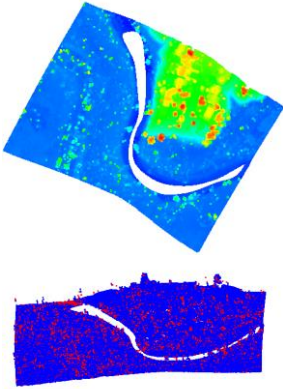
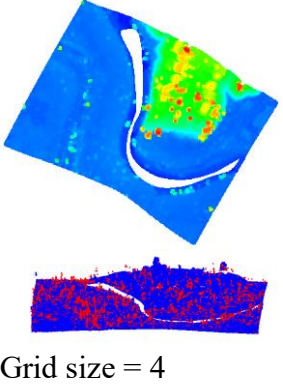
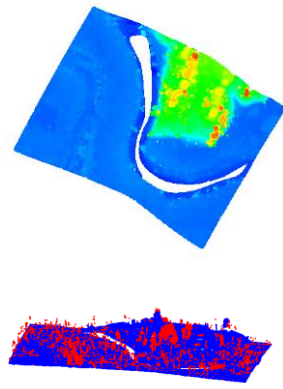
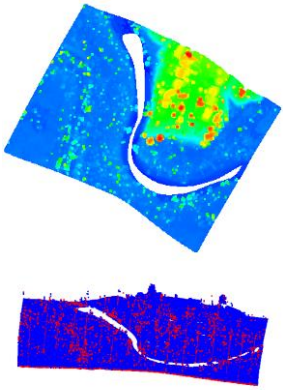
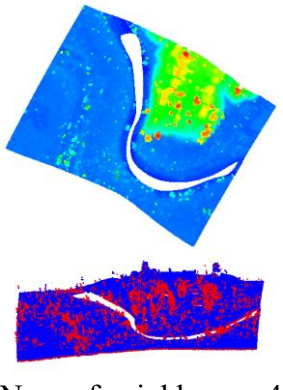
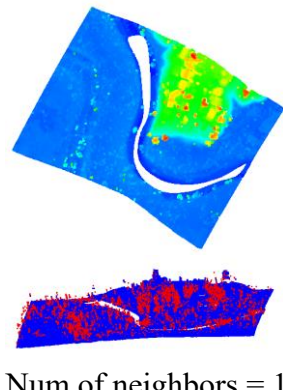


Figure 1: Original image without denoising

The following table compares the intensity in 2D and classification in 3D of different denoising methods and sampling sizes.

	Sampling Size		
Denoising method	Small	Medium	Large
Density based			
	Radius = 0.25m	Radius = 1m	Radius = 4m

Grid based density	 Grid size = 1	 Grid size = 4	 Grid size = 16
Distance to neighbor	 Num of neighbors = 1	 Num of neighbors = 4	 Num of neighbors = 16

Analysis and explanation of choice

To select the optimal result from various denoising methods, the criteria should be defined first. It is suggested that an optimal result should maintain a balance between removing outliers and preserving fine detail, such as the ground features while low in residual noise.

From the table, it appears that low and medium sampling size tends to leave too much residual noise. This is especially visible in the 3D classification view, whereas for large sampling size the algorithm removes the noise sufficiently aggressive. Therefore, it appears that large sampling size in general, across denoising methods, is the optimal choice for denoising.

The three methods behave differently on ALS point clouds:

Density-based (Radius 0.25 m / 1 m / 4 m) removes points below a local density threshold. While it retains some clustered features, it leaves many isolated outliers, especially at medium and large radii, as visible in the 3D classification views.

Grid-based density (Grid size 1 / 4 / 16) subdivides the cloud into regular cells and eliminates low-density cells. This method is aggressive and successfully removes isolated points, but it also

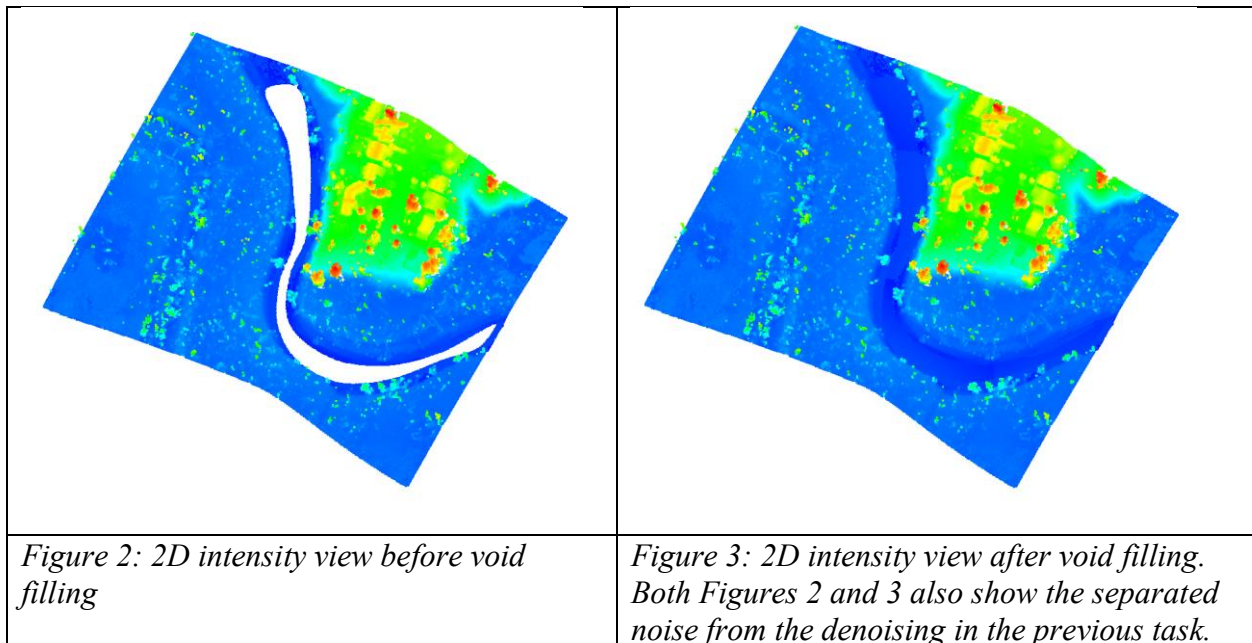
deletes small legitimate clusters (e.g., ground features in sparse scan areas), leading to over-smoothing.

Distance to neighbor (Num of neighbors 1 / 4 / 16) classifies a point as noise if it has fewer than the specified neighbors within the search radius. This method maintains the best balance: it efficiently eliminates truly isolated outliers while preserving small clustered features and ground detail.

Therefore, it appears that the distance to neighbor method with 16 neighbors achieves the optimal result, by having a balance between preserving small clustered features and removing isolated outliers.

Task 2 – Void filling and hydroflattening

Void filling



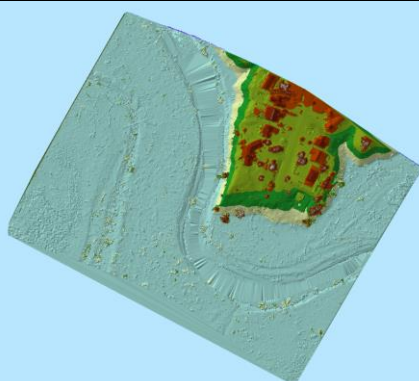
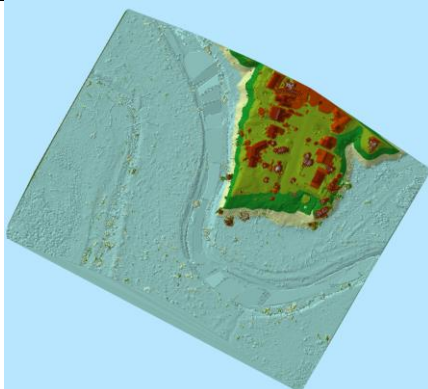
Data points before void filling						Data points after void filling					
ID /	Class Name	Size				ID /	Class Name	Size			
0		753,086				0		753,086			
1		183,446				1		183,446			
2		23,757				2		23,757			
Total points: 936 532						Total points: 960 289					
Where class 0 is the original points, class 1 is the points removed by denoising. The noise											

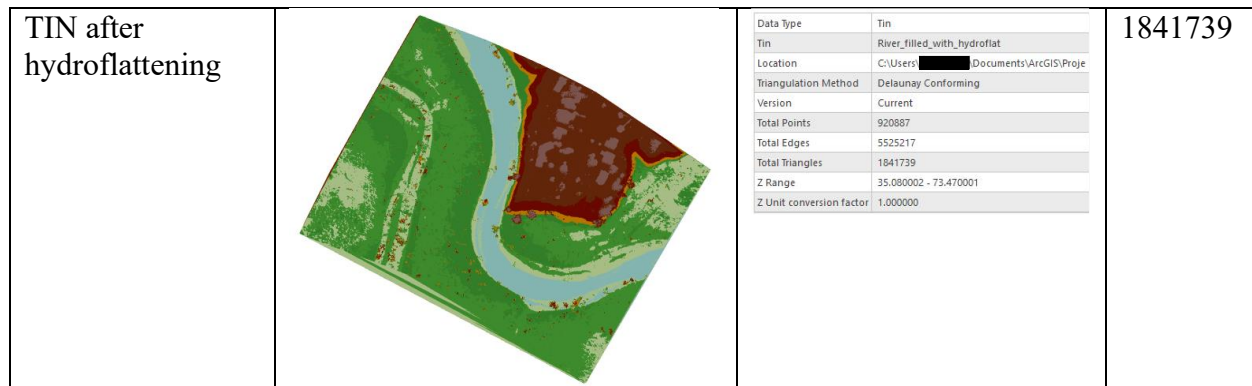
points are also included in the 2D intensity view.	Includes the classes before void filling and class 2, which is the filled points.
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Void filling in 3DMas interpolates gaps (waterbodies in this exercise) using surrounding valid points. Before filling, the dataset contained 936 532 points (including class 0 original points and class 1 noise separated by denoising). After filling, the total number of points to 960 289 with the addition of class 2 interpolated points. This expected increase occurs because the software generates new points to create a continuous surface, improving DEM continuity without altering the original measured points.

Hydroflattening

Hydroflattening enforces constant elevation on water surfaces (lakes, rivers, reservoirs) to follow the typical topographic DEM behavior, where water appears flat with homogeneous height. This prevents TIN assigning fluctuating heights to water areas. Without hydroflattening, TIN interpolation over water voids, especially filled by section, may create “steps,” due to irregular heights. They appear as jagged artifacts and do not meet the topographic principles where water surfaces tend to be flat.

Stage	TIN Screenshot	TIN properties source screenshot	Triangle count																				
TIN from original point cloud data		<table><tr><td>Data Type</td><td>Tin</td></tr><tr><td>Tin</td><td>River_LasDatasetToTin</td></tr><tr><td>Location</td><td>C:\Users\ [redacted] \Documents\ArcGIS\Proje</td></tr><tr><td>Triangulation Method</td><td>Delaunay Conforming</td></tr><tr><td>Version</td><td>Current</td></tr><tr><td>Total Points</td><td>896090</td></tr><tr><td>Total Edges</td><td>5376432</td></tr><tr><td>Total Triangles</td><td>1792144</td></tr><tr><td>Z Range</td><td>35.080002 - 73.470001</td></tr><tr><td>Z Unit conversion factor</td><td>1.000000</td></tr></table>	Data Type	Tin	Tin	River_LasDatasetToTin	Location	C:\Users\ [redacted] \Documents\ArcGIS\Proje	Triangulation Method	Delaunay Conforming	Version	Current	Total Points	896090	Total Edges	5376432	Total Triangles	1792144	Z Range	35.080002 - 73.470001	Z Unit conversion factor	1.000000	1792144
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Z Unit conversion factor	1.000000																						
TIN after void filling, no hydroflattening		<table><tr><td>Data Type</td><td>Tin</td></tr><tr><td>Tin</td><td>River_filled3_LasDatasetToTi</td></tr><tr><td>Location</td><td>C:\Users\ [redacted] \Documents\ArcGIS\Proje</td></tr><tr><td>Triangulation Method</td><td>Delaunay Conforming</td></tr><tr><td>Version</td><td>Current</td></tr><tr><td>Total Points</td><td>919713</td></tr><tr><td>Total Edges</td><td>5518167</td></tr><tr><td>Total Triangles</td><td>1839389</td></tr><tr><td>Z Range</td><td>35.080002 - 73.470001</td></tr><tr><td>Z Unit conversion factor</td><td>1.000000</td></tr></table>	Data Type	Tin	Tin	River_filled3_LasDatasetToTi	Location	C:\Users\ [redacted] \Documents\ArcGIS\Proje	Triangulation Method	Delaunay Conforming	Version	Current	Total Points	919713	Total Edges	5518167	Total Triangles	1839389	Z Range	35.080002 - 73.470001	Z Unit conversion factor	1.000000	1839389
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After void filling, the number of triangles increased from 1792144 to 1839389, around 3% increase. Since the number of points increases, the data becomes more complex, therefore the number of triangles increased probably reflected the added interpolated points. However, due to the new points having a relatively homogeneous height, it is likely due to the fact that fewer triangles are needed to store them, causing a modest increase.

After hydroflattening, the number of triangles also increased. The progressive yet modest increase in triangles reflects the additional refinement such as void filling and hydroflattening. From the above screenshots, before hydroflattening the TIN shows visible inconsistencies on the water surface, whereas post-hydroflattening the TIN appears to be smooth, the water surfaces have clean edges, and it is more suitable for representing hydrological reality.

Conclusion

This lab assignment successfully transformed a noisy, incomplete ALS point cloud into a high-quality, hydroflattened DEM suitable for topographic and hydrological applications. For the denoising part, the distance to neighbor (16 neighbors) was selected for outlier removal while preserving important ground detail. Void filling eliminated the data gaps. Hydroflattening addressed the issues of water surface artifacts. Overall, the workflow reduced noise using the optimal results and increased data completeness, with only a small increase in triangle count.

Limitations include using the manual polygon selection tool to fill the voids and trim the filling. Future improvements could involve semi-supervised methods such as one-click segmentation, or even unsupervised filling using automated parameter optimization or machine-learning-based selection and filling. For denoising, balancing aggressiveness and detail preservation could be a challenge, and this exercise selected one method out of the 9 denoising results. This exercise highlights the importance of pre-processing laser scanning point cloud data.

Overall, this lab assignment highlights the importance of preprocessing in laser scanning, such as denoising, void filling, and hydrological correction to make the point clouds more reliable and matches the real world topography for geospatial use.